Ayomide Aisida

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Education

Bowie State University *Expected Grad: Dec 2027*

- *Bachelor of Science in Computer Science | GPA:* 4.0
- **Relevant Coursework:** Artificial Intelligence, Data Science, Data Structures & Algorithms, Design & Analysis of Algorithms, Database Management, Operating Systems, System Programming

Certifications

Google AI Essentials (Google), Programming Methodologies (Code in Place, Stanford University)

Experience

Teaching Assistant *Aug 2026 – Present*

Computer Science Department, Bowie State University

- Support students in Data Structures & Algorithms, Database Management, and System Programming courses through weekly tutoring, technical guidance, and problem-solving.
- Assist professors during class by answering questions, explaining technical concepts, and supporting students with course exercises and projects.

AI/ML Research Intern *May 2026 – Jul 2026*

Summer AI Research Institute, Morgan State University (CEAMLS)

- Integrated 6 public dermatology datasets with inconsistent schemas and demographic fields into a unified 10,250-image benchmark; cleaned data, handled missing values, and standardized labels into 8 diagnosis classes.
- Benchmarked 5 deep learning architectures (Swin Transformer, ConvNeXt, EfficientNetV2-M, EfficientNet-B0, ResNet50) for skin lesion classification under an identical training pipeline, achieving up to 76.7% test accuracy.
- Evaluated model fairness across skin-tone groups using subgroup recall and minimum-group recall, applying Grad-CAM and SHAP to interpret predictions and identify potential bias patterns.

Undergraduate Research Assistant *May 2025 – Jun 2026*

Cypher Lab, Bowie State University

- Developed Python/Flask backend and database components for S.P.A.R.K.I., an AI-powered coding education platform, using SQLAlchemy and MongoDB to support learner progress tracking and personalized content delivery.
- Integrated LLMs for personalized feedback, real-time hint generation, and automated assessments, and implemented an adaptive recommendation algorithm using hash-based data structures for O(1) user-profile retrieval.
- Prototyped scalable LLM infrastructure using Docker, vLLM, Redis, API gateways, and load balancing to support distributed model inference.

Projects

DermaBridgeAI | *TypeScript, JavaScript, Expo, React Native, Python*

- Built an AI-powered skin cancer screening app using a fairness-first multimodal pipeline to classify lesion images and support equitable performance across diverse skin tones (Fitzpatrick I–VI).
- Led the end-to-end DermaBridgeAI workflow by building the mobile scan intake, FastAPI inference pipeline, and referral flow, integrating Expo Router, TypeScript, and Python services to deliver lesion screening and dermatologist follow-up.

Smart Meeting AI Assistant | Node.js, Express, IBM watsonx.ai

- Co-built a Smart Meeting AI Assistant at the IBM Hackathon, leading backend development to capture meeting decisions and automate action item tracking using IBM Watson Orchestrate agents.
- Built 6 backend REST APIs to integrate IBM Watson Orchestrate agents with application services, enabling real-time action-item extraction and automated task assignment from meeting transcripts.

Hotel Booking Management System | Java, UML

- Designed and built a Java hotel booking application with reservation and customer management features, leveraging OOP, binary search, sorting, and hash maps for efficient data handling.

Technical Skills

Languages: Python, Java, C, C++, JavaScript/TypeScript, SQL, HTML/CSS
AI/ML: PyTorch, TensorFlow/Keras, scikit-learn, vLLM, CNNs, Vision Transformers, Grad-CAM, SHAP, Prompt Engineering
Frameworks & Tools: FastAPI, Flask, React, Node.js, Express.js, REST APIs, Docker, Git/GitHub, Linux, Redis, SQLAlchemy

Leadership & Honors

Leadership: Research Ambassador; Chair of Logistics, Women in Computer Science (WICS); Vice President, Bulldog Coders
Awards: Dean's List, TMCF Scholar
Competitive Programming: ICPC Mid-Atlantic 2025, CCSC Eastern Regional Programing Contest 2025
Affiliations: CodePath, ColorStack, Rewriting The Code, Girls Who Code